

Andre C.A. de Carvalho

Full-Stack Developer

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Full-Stack Developer with 7+ years shipping web and mobile products in production, with hands-on AI integration experience (Vision LLM in a production OCR pipeline). Solid foundation in Go/Python/TypeScript, microservices (gRPC, NATS), marketplace/payment integrations, and a strong data background (ETL, SQL, Python).

- **Backend:** Go · Python · gRPC · NATS · Microservices
- **Frontend & Mobile:** React · Next.js · TypeScript · Flutter
- **Marketplaces:** Mercado Livre · Amazon · Shopee
- **Payments:** PagarMe · PinPag
- **AI Integration:** Vision LLM (via API) · OCR pipeline · Ollama · MCP

EXPERIENCE

Full-Stack Engineer Jan 2025 – Present

Beezoo Labs — Multi-tenant SaaS platform for loyalty and digital commerce (malls & retail)

- Built from scratch the Showcase microservice (digital storefronts with QR codes, pricing policies, and catalog enrichment via NATS events) plus its mobile-first frontend — end-to-end feature: .proto → Go service → admin panel → public app.
- Led the app-whitelabel initiative in the Go microservices monorepo (gRPC, NATS JetStream, Outbox Pattern, PostgreSQL, KrakenD), enabling multi-tier mobile app configuration per tenant.
- Developed white-label Flutter apps for consumers and partners — loyalty, campaigns, Eventim integration, biometric login, and unified design system.
- Solo-built an OCR service for Brazilian tax receipts (NF-e/NFC-e) in Python/gRPC integrating existing tools in a cascade (QR → Tesseract → PaddleOCR → Vision LLM via API), raising accuracy from 28% to 75% on a 1,138 real-image dataset; deployed on Docker + Kubernetes.

Go TypeScript Dart Python gRPC Protobuf NATS PostgreSQL Next.js 15 React 19 Flutter Docker
Kubernetes Tilt

Software Engineer May 2022 – Dec 2024

Hubsell — E-commerce Integration Platform

- Built frontend and integrations for Seller Center/Dashboard, connecting sellers to marketplaces and external APIs.
- Integrated payment gateways (PagarMe, PinPag) and evolved critical product flows.
- Shipped conversion-impacting features (POS and Shopping Assistant).
- Improved UX/performance on heavy screens using ElasticSearch + React Query.

React Next.js Node.js MongoDB ElasticSearch Zustand React Query Expo

Full-Stack Developer 2020 – 2022

3 UP Tech — 3Ponto Financial App

- Built and maintained web and mobile apps end-to-end.
- Backend development using a microservices architecture and legacy integrations.
- Led frontend initiatives: feature delivery, bug fixes, and performance improvements.

React React Native Node.js PostgreSQL Express Microservices

Upwork — Data & Business Systems

- Built ETL pipelines and Power BI dashboards for help desk analytics.
- Implemented ERP workflows (Odoo/OpenERP) with custom business processes.
- Automated data models with SQL and Python for executive dashboards.

Python SQL PostgreSQL MySQL Odoo Power BI

SKILLS

AI Integration

Production LLM integration · Vision LLM via API · OCR pipeline · Ollama · MCP

Frontend

React · Next.js · TypeScript · State (Zustand/Redux) · Storybook

Backend

Go · Node.js · Python · gRPC · REST APIs · NATS · Microservices

Data & Infra

PostgreSQL · MongoDB · Redis · Docker · Kubernetes · ElasticSearch · ETL/SQL

Quality

Jest · Vitest · Playwright · CI/CD

EDUCATION

- Postgraduate studies in Business Intelligence (not completed) — Unicesumar, 2017–2018
- B.B.A. in Business Administration — União de Faculdades Metropolitanas de Maringá, 2016

PERSONAL PROJECT

SEIF Protocol — Research on provenance of AI-generated artifacts

2026 – Present

Personal project · seifprotocol.com

- Personal research project on cryptographic provenance (Ed25519, optional time-anchoring via OpenTimestamps) and auditable governance for AI-generated artifacts.
- Motivated an academic publication on Zenodo (see Publications).

PUBLICATIONS

- CARVALHO, A. C. A. de. **On the Unique Primitive Second-Order System with Golden-Ratio Damping: Mathematical Properties and Applications to AI Context Verification.** Zenodo, 2026. DOI: [10.5281/zenodo.19344678](https://doi.org/10.5281/zenodo.19344678).